

Name and surname: _____ Student number: _____

UNIVERSITY OF CAPE TOWN
DEPARTMENT OF STATISTICAL SCIENCES - STA2020F: APPLIED STATISTICS
CLASS TEST 3 – 15 May 2026

Time: 100
MINUTES

Max Marks: 40

Internal examiners: Mrs. D. van Wyk-de
Ridder
Ms. A Cloete

Instructions

Answer ALL questions in the answer blocks provided. You may **CIRCLE** chosen letter options for multiple-choice and multiple-select type questions.

Please note that you will need access to **RStudio** to complete this test. You need to be prepared to import a dataset in .csv format into R and to use the basic commands introduced during your lectures and the RStudio lab sessions/pictorials. You will also be required to answer questions about the content covered in Weeks 7 to 10, as well as R labs of Weeks 8 to 11, of the course and perform calculations by hand.

NB: Round all numeric answers to **TWO DECIMAL PLACES**, unless instructed otherwise.

Where necessary, use the ***as.factor()*** function in R to transform variables to factors.

Make sure to use "." as a **decimal point** and not a comma.

If necessary, increase the level of precision of R output by running: ***options(digits=12)*** before you do any other commands.

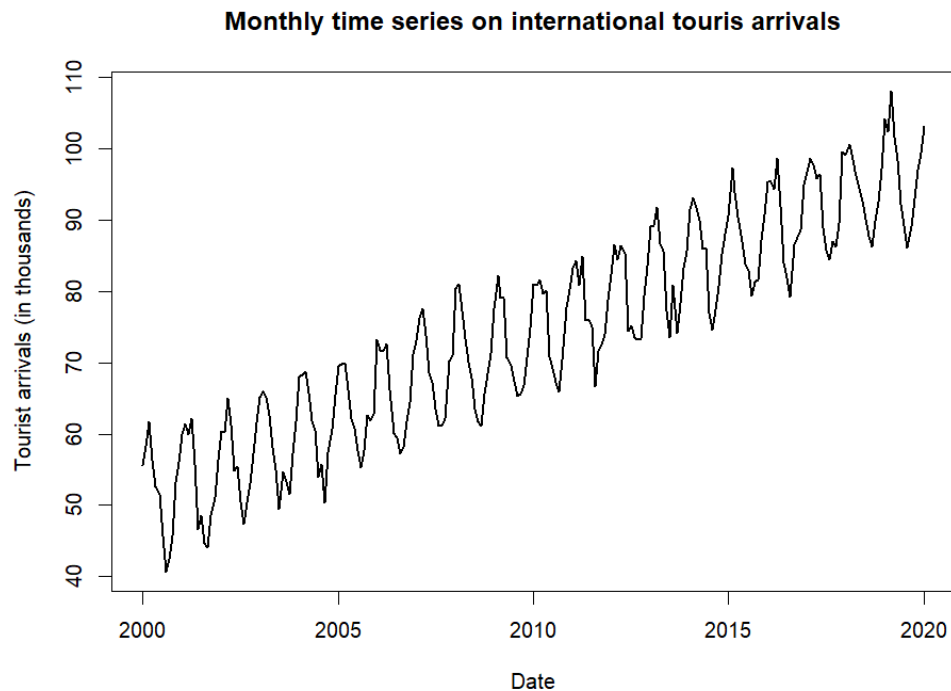
Follow the section-specific instructions communicated in the different parts of this test.

The following files will be downloaded for you by an invigilator. These are the only resources, together with RStudio, that you are allowed to use for this test.

GENERAL INSTRUCTIONS
RSTUDIO CHEAT SHEET
FORMULA SHEET
STATISTICAL TABLES
DATASET: *mixtapes.csv*
DATASET: *puzzle_strategies.csv*
DATASET: *study_techniques.csv*

Good luck!

A travel agency investigates the monthly number of international tourist arrivals to a coastal destination between January 2000 and January 2020. Tourism demand has steadily increased over the period due to improvements in infrastructure, destination marketing, and growth in international travel. The time series is displayed below.



The following table contains some of the time-series observations on international travellers. Moving average smoothing has been applied. Some values are left out and need to be calculated.

Date	t	Y_t	MA(12)	CMA(12)
Apr 2002	28	61.33	-	-
May 2002	29	54.93	-	-
Jun 2002	30	55.51	-	-
Jul 2002	31	50.45	56.87	-
Aug 2002	32	47.38	57.34	A
Sept 2002	33	50.39	57.35	-
Oct 2002	34	53.13	-	-
Nov 2002	35	57.24	B	-

Dec 2002	36	61.58	-	-
Jan 2003	37	65.11	-	-
Feb 2003	38	65.90	-	-
Mar 2003	39	65.19	-	-
Apr 2003	40	62.67	-	-
May 2003	41	58.21	-	-
Jun 2003	42	54.98	-	-

Suppose the time series on international travellers is now limited to 2012-2016, i.e., our time series consists of only 5 years' worth of data, and you may disregard the rest. The detrended data (sorted by month) for the five years under consideration are shown in the table below. Some values are left out and need to be calculated.

	January	...	December	Total
2012	0.9598	...	0.9761	
2013	0.9968	...	1.0131	
2014	0.9838	...	1.0202	
2015	0.9619	...	1.0033	
2016	0.9811	...	1.0468	
Seasonal Index	C	...	1.0119	12.4885
Adjusted Seasonal Index			D	

1. Based on the time series plot of international tourist arrivals, which of the following statements best describes the time series? Circle the correct letter. [2]
- A. The time series exhibits an increasing linear trend with no seasonal pattern.
 - B. The time series is stationary because its variance is constant.
 - C. The time series exhibits an increasing linear trend with strong seasonality.**
 - D. The time series exhibits an increasing linear trend with regular instances of cyclicity but no seasonality.
 - E. The time series exhibits cyclical fluctuations and seasonality with an increasing linear trend.

2. The following question pertains to time series data in general and not necessarily the time series on international tourist arrivals. Which of the following statements about time series data are generally true? Circle all the letters that apply. [2]

- A. The presence of autocorrelation indicates that a time series has a constant mean.
- B. Serial correlation refers to the correlation between successive observations in a time series.**
- C. Time series observations are collected at equally spaced time intervals.**
- D. The order of the observations in a time series can be rearranged without affecting the analysis.
- E. Most time series do not have random variation.

3. Indicate whether the following statement is TRUE or FALSE by writing either 'T' or 'F':
We would be interested in applying moving-average smoothing to the time series of international travellers to improve forecasts of future values. [1]

False: Moving-average smoothing is not used for forecasting.

4. Indicate whether the following statement is TRUE or FALSE by writing either 'T' or 'F':
The adjusted seasonal index for October is calculated as $S_{10} = 0.925$. This implies that, on average, the number of international travellers in October is 0.075 times lower than the annual average. [1]

False; it implies that, on average, the number of international travellers in October is 7.5% lower than the annual average.

5. What is the value of **A**, the centred moving average value associated with August 2002? Only your final answer will be marked. [1]

$$\frac{56.87 + 57.34}{2} = 57.105 = 57.11$$

6. What is the value of **B**, the moving average value associated with November 2002? Only your final answer will be marked. [1]

$$\frac{55.51 + \dots + 58.21}{12} = 57.73$$

7. In classical decomposition, why is the original time series often deseasonalised before analysing the trend component? Limit your answer to one sentence. [2]

The removal of seasonal variations results in a smoother or seasonally adjusted (1) time series, making it easier to estimate the trend component (1).

8. Which condition of stationarity is most likely violated in the time series of international travellers? Limit your answer to one sentence. [1]

The mean is not constant over time (1). No mark for mentioning variation – the variation seems constant, and the condition is not obviously violated.

9. What is the value of **C**, the seasonal index for January? Only your final answer will be marked. [1]

$$\frac{0.9598 + \dots + 0.9811}{5} = 0.9767$$

10. What is the value of **D**, the adjusted seasonal index for December? Only your final answer will be marked. [1]

$$\frac{1.0119 \times 12}{12.4885} = 0.9723$$

Time Series: Section 2

[13]

The time-series dataset in the file *mixtapes.csv* captures quarterly sales numbers for mixtapes from June 1990 to June 2015.

Instructions:

- i)** Load the following packages: *tidyverse*, *ggplot2*, *fpp2*, *forecast*, *urca*
 - ii)** Import the file *mixtapes.csv* into R as an object called *my.data*. Do not open or edit the file in Excel.
 - iii)** Convert the imported values into a time series by using the following command exactly as typed here: ***my.ts <- ts(my.data\$value, start = c(1990, 2), frequency = 4)***. This time series will henceforth be referred to as the original time series.
 - iv)** Plot the time series.
 - v)** Construct and interpret the correlogram (ACF) for the original series.
 - vi)** Use a Box-Cox transformation **and** first-order differencing to remove trend and/or seasonality (as deemed applicable) from the original time series and examine the residuals of your transformed series (no model fitted, just the residuals of the transformed time series).
 - vii)** Split the raw time series into a training set from the first observation in 1990 until the last observation of 2011, and a test set containing all remaining observations.
- Fit:**
- (1) the naïve method and
 - (2) the most appropriate long-term simple forecasting method for the training data, and compare their **forecast accuracy** on the test data. Make sure to set your forecasting horizon to the total number of observations in your test dataset.

11. Consider instruction (iv). Based on the time series plot of sales numbers for mixtapes, which of the following statements best describes the series? Circle the correct letter. [2]

- A. The time series exhibits a decreasing linear trend with seasonality and random variation.
- B. The time series shows an increasing linear trend with no seasonality.
- C. The time series exhibits a decreasing linear trend with random variation, but no clear seasonality or cyclicity.**
- D. The time series exhibits an increasing linear trend with multiple instances of cyclicity.
- E. The time series shows a decreasing linear trend with four clearly discernible seasons in a year.

12. Indicate whether the following statement is TRUE or FALSE by writing either 'T' or 'F':
Based on the time series plot generated in instruction (iv), the most appropriate simple forecasting method among the standard simple forecasting methods for long-term forecasting is the naïve method. [1]

False - the most appropriate method for long-term forecasting is drift, as it accounts for the trend.

13. Which of the following statements is incorrect? Consider instructions (v) and (vi) and circle all the letters that apply. [2]

- A. The autocorrelation plot of the original time series data shows a statistically significant positive relationship for observations 10 time periods apart.
- B. A square root transformation is necessary to stabilise the variance of this time series.**
- C. First-order differencing at lag 4 will detrend the original time series.**
- D. Only when the variance is stable, and the mean of the time series is zero, can any time series be assumed to be stationary.**
- E. If significant autocorrelation remains in the residuals of the original time series after the variance has been stabilised and first-order differencing has been applied, it means that there is still information left to uncover in the time series.

14. After performing instruction (vi), investigate the R output on the residuals of your transformed time series. Make a conclusion about this test. Limit your answer to a single sentence. [1]

We conclude that autocorrelation is present in at least one of the considered lags (1).

15. Is the correlogram of the residuals of the transformed time series from instruction (vi) indicative of a white noise process? Answer in one sentence and explain your reasoning.

[2]

No (1), the correlogram shows significant autocorrelation (0.5) at multiple lags (0.5) and can therefore not be from a white noise process.

16. Carry out instruction (vii). What is the MSE of the most accurate simple forecasting model, based on your R output? Only your final answer will be marked.

[2]

$$2.630334^2 = 6.92$$

17. Carry out instruction (vii). Indicate whether the following statement is TRUE or FALSE by writing either 'T' or 'F': *Based on your R output, the largest mean absolute error is associated with the naïve forecasting method.*

[1]

True - naïve MAE = 5.99 > 2.13 drift MAE

18. Suppose the last observation of the training dataset is 349.9937, and the residual standard deviation of the naïve method is 2.6437. What is the upper bound of the 95% prediction interval for the next sales number for mixtapes (the first observation of the test data set) when using the naïve method? Only your final answer will be marked.

[2]

$$349.9937 + 1.96 * 2.6437 = 355.18$$

Non-parametric tests: General

[2]

19. A sports science researcher at the University of Cape Town wants to compare the effectiveness of three different recovery methods on athletes' perceived muscle soreness after intense training sessions. The three recovery methods are:

- Ice bath
- Compression garments
- Active recovery (light cycling)

Twelve athletes participate in the study. Each athlete completes three identical high-intensity training sessions, spaced one week apart to avoid carryover effects. After each session, they are randomly assigned one of the three recovery methods, ensuring that every athlete experiences all three methods over the course of the study.

Twenty-four hours after each session, athletes rate their muscle soreness on a scale from 1 (no soreness) to 10 (extreme soreness). Which non-parametric test is appropriate for this analysis? Circle the correct answer. [2]

- A. Wilcoxon Rank Sum (Mann–Whitney U) test
- B. Wilcoxon Signed Rank test
- C. Kruskal–Wallis test
- D. Spearman's rank correlation
- E. Friedman test**

Non-parametric tests: Scenario 1

[5]

A teacher believes that a new active-recall study technique improves performance on weekly biology quizzes. She randomly selects 8 students. Each student writes a baseline quiz BEFORE, completes two weeks of training in the technique, and then takes a post-training quiz of equivalent difficulty AFTER. The data are stored in *study_technique.csv*. Inspection of the differences shows clear right-skew, so a paired t-test is not appropriate.

20. Using the data in *study_technique.csv*, manually code the test statistic *W* in RStudio.

Show the R code you used and enter the value of *W*.

[2]

Answer:

```
> recall <- read.csv("study_technique.csv")
> recall$diff <- recall$Before - recall$After
> recall$abs_diff <- abs(recall$diff)
> recall$rank <- rank(recall$abs_diff)
> recall$signrank <- recall$rank * sign(recall$diff)
> sum(recall$signrank)
[1] -34
```

- If the correct answer of -34 is given without anything else: half a mark
- More than one missing step or error in code: half a mark
- Full marks only for correct code and correct answer.

21. For the following question use, *wilcox.test()* to conduct the test and make sure to answer each of the three parts: [3]

- Based on the output, state the null and alternative hypotheses (in words, and without referring to ANY symbols, i.e., not typing 'mu' to represent a mean, etc.).
- Using a 5% significance level, decide whether to reject or not reject the null hypothesis by supplying evidence for your decision, and
- Interpret the decision in the context of the experiment.

Mark with error: If a student gets the decision wrong but their conclusion is consistent with their (incorrect) decision, award the mark for the conclusion.

Marker's note: No marks for clearly wrong hypotheses.

Model answer:

Null hypothesis: There is no difference in the performance of students before and after the introduction of the new recall technique. [0.5]

Alternative hypothesis: Students perform better after the introduction of the new recall technique. [0.5]

Based on the output of *wilcox.test()*, the p-value is less than 0.05 [0.5], so the null hypothesis is rejected. [0.5]

This provides evidence that the new recall technique improves performance. [1]

Non-parametric tests: Scenario 2

[5]

An educational psychologist compares four problem-solving strategies (A, B, C, D) for solving a Sudoku puzzle. She randomly assigns 32 participants to one of the four strategies (8 per group) and records the time (in minutes) each participant takes to solve the puzzle. The Q-Q plots are clearly non-normal in two of the four groups, so a one-way ANOVA is not appropriate. The data are stored in *puzzle_strategies.csv*.

22. State the null and alternative hypotheses for the appropriate non-parametric test (in words, and without referring to ANY symbols, i.e., not typing 'mu' to represent a mean, etc.) [1]

Null: There is no difference in median time taken to solve the puzzle across the strategies. [0.5] / There is no difference in time taken to solve the puzzle across the strategies.

Alternative: There is a difference in the median time taken to solve the puzzles across the strategies. [0.5] / There is a difference in time taken to solve the puzzle across the strategies. [0.5]

23. Using R, compute the test statistic manually (i.e. without any built-in functions). Provide the sum of the rank sums for each group. In other words: [2]

$$\sum_{i=1}^k T_i$$

Only your final answer will be marked.

582

24. Use the appropriate built-in function in R and report the test statistic from the output. Only your final answer will be marked. [1]

21.50

Non-parametric tests: Scenario 3

[3]

A campus health study collects daily screen-time (in hours) and self-reported sleep quality (on a 1–10 scale) from 35 students over a one-week period. She suspects that increased screen time is associated with a decrease in sleep quality. Both variables are skewed and ordinal-like, so the researcher decides to use a Spearman correlation coefficient to test her suspicions. The following R output is obtained:

```
Spearman's rank correlation rho

data: sleep$SleepQuality and sleep$ScreenHours
S = 11057, p-value = 0.000322
alternative hypothesis: true rho is less than 0
sample estimates:
    rho 
-0.5486131
```

25. Indicate whether the following statement is TRUE or FALSE by writing either 'T' or 'F':

The following R code was used to produce the output shown:

```
cor.test(sleep$SleepQuality, sleep$ScreenHours, method = "spearman", alternative = "less")
```

[1]

True

26. Given the p-value and the value of r_s as shown on the previous page, circle *all* letters pertaining to correct interpretations at the 5% significance level. [2]

- A. There is a moderate negative monotonic association between screen time and sleep quality.**
- B. Higher screen time causes lower sleep quality in this population.
- C. The correlation is statistically significant ($p < 0.05$).**
- D. Spearman's r_s captures only the linear component of the association, like Pearson's r .
- E. Students who spend more hours on screens tend to report lower sleep quality.**